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Class 9
Chemistry
Chapter 7 Electrochemistry
NOTES

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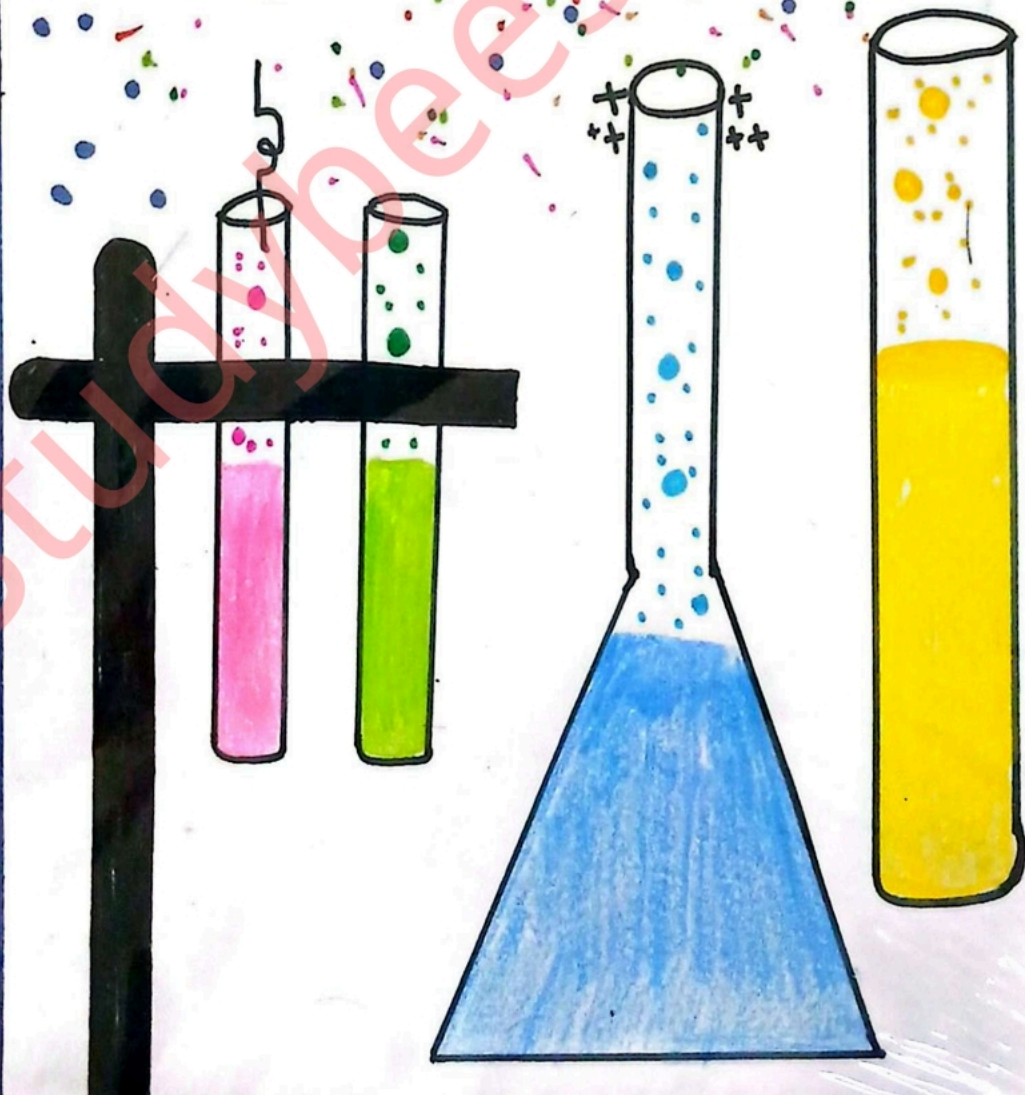
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CLASS 9

CHEMISTRY



Unit no 7:~ Electrochemistry

Topic:~

7.1

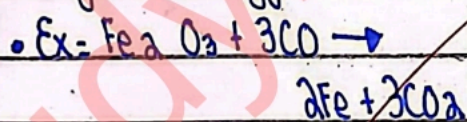
"Oxidation & reduction"

Question no 1:- differ oxidation & reduction with example.

Answer:~

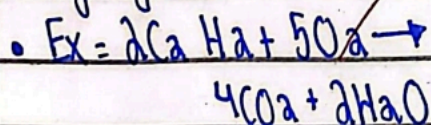
Oxidation

- Oxidation is defined as "gain of oxygen".



"CO" is gaining O so it's under going oxidation

- Oxidation is the "Loss of hydrogen".



"CaH₂" is losing H so under going oxidation.

Reduction

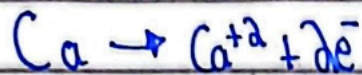
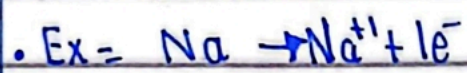
- Reduction is defined as "Loss of oxygen".

- Fe + O₂ is losing O, So it is under going reduction.

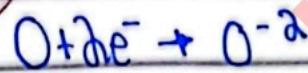
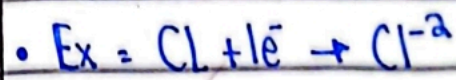
- Reduction is the "gain of hydrogen".

- "O₂" is going hydrogen So under going reduction

• It is defined as
"Lose of electron"



• It is defined as "gain of
electron"



EXAMPLE 7.1 :-

Question :-

Following the action occurs when you burn the gas
 $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + \text{Heat}$

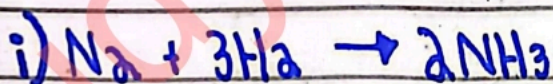
Identify the element undergoing oxidation.

Answer :-

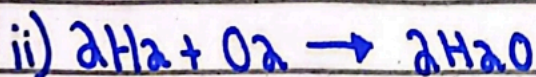
Here 'C' in CH_4 loses hydrogen and combines with O-atoms, thus C-atom undergoes oxidation. At the same time O-atoms combine with H-atom from H_2O . Thus, O-atom undergoes reduction.

Concept Assessment 7.1

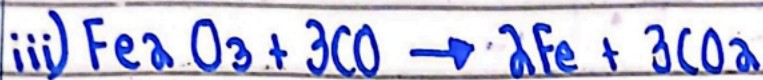
Identify the element undergoing oxidation & reduction in the following reactions.



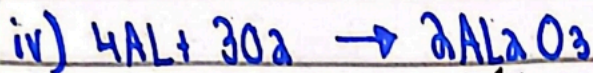
"N"-atom combines with "H"-atom so undergoes reduction whereas H-atom is oxidation.



H-atom gains O-atom so undergoes oxidation & O-gains H & so undergoes reduction.



Fe-atom Lose O-atom So undergo reduction
whereas, C-gain O & is oxidation.



Al-atom gain O-atom So undergo oxidation & at
the same O-atom is reduced.

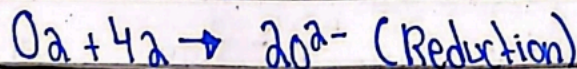
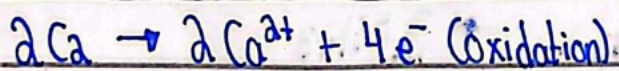
EXAMPLE:- 7.2

▶ In the following reaction identify which element is
reduction & oxidation.



Answer:-

Ca belongs to group IIA in the P.T. It loses $2e^-$ to form
 Ca^{2+} (Oxidation) whereas O belong to group VIA It
gain $2e^-$ to form O^{2-} (Reduction)

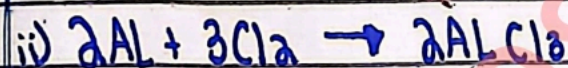
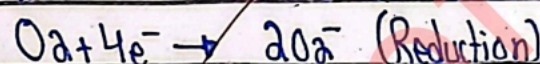
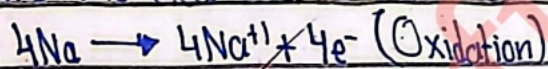


Concept Assessment 7.2

In the following reactions identify which element is oxidized & reduction in term of electron transfer.



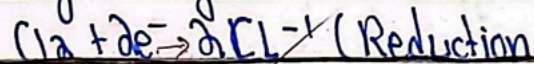
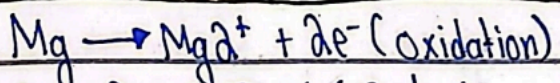
Na belong to IA, each Na-atom loses an electron to form Na^+ (Oxidation) whereas, O belong to group VIA each O-atom gains two electrons to form O^{2-} (Reduction)



Al belong to group IIIA, each Al-atom loses 3-electron to form Al^{+3} (Oxidation), Cl belong to group VIIA, each Cl atom gain an electron to form Cl^- (Reduction)



Mg belong to group II-A, It loses two electron to form Mg^{2+} (Oxidation), Cl belong to group VII-A each Cl-atom gains an e^- to form Cl^- (Reduction)



Topic:

7.2

"Oxidation state & rule for assigning oxidation state"

Question no 1:- Define oxidation State or oxidation number.

Answer

It is defined as the number of change of an atom will have in a molecule or a compound.

The element that shows increase in oxidation no is oxidation & the element that decrease in oxidation no is reduced.

Question no 2:- What are the rules of assigning oxidation State or number?

Answer:-

i) The oxidation State of any uncombined or free element is always Zero.

eg:- S in S_8 is Zero H in H_2 is Zero

ii) In Simple ion's oxidation State of atoms is equal to the charge on their e.g.: oxidation State of Na in Na^+ & Ca in Ca^{2+} is +1 & +2 respectively.

iii) In a Complex ion, Oxidation State is equal to to charge of their ion.

e.g.:~ in NH_4^+ the oxidation State of N & 4H is 1

iv) The oxidation of each of the atoms in a molecule's compound is counted separately & their algebraic sum is Zero.

e.g.:~ in CO_2 , oxidation state of "1" C & "2" O is Zero.

Question no 3:- What are oxidation no of mono-atomic ions.

Answer:-

The oxidation number of mono atomic ions are equal to their charge.

e.g.:~

Na^+ is formed after a Na-atom has lost $1e^-$ to gain a +1 charge so its O.N is +1.

Question no 4:- What are oxidation number of polyatomic ions.

Answer:-

In polyatomic-ion, the sum of the oxidation state is equal to the charge on ions.

e.g.:~ CO_3^{2-} ion, the O.N of C & O are +4 & -2 respectively, so the sum of the O.N of 1-C & 3-O would be $1(+4) + 3(-2) = -2$ which is the charge on ion.

EXAMPLE 7.3

Determine oxidation no of Cr in
 $K_2Cr_2O_7$

Solution:-

let $n = \text{O.N. of Cr}$

$$\text{then } 2(+1) + 2n + 7(-2) = 0$$

$$+2 + 2n - 14 = 0$$

$$2n - 12 = 0$$

$$2n = 12$$

$$n = 12/2$$

$$n = 6$$

EXAMPLE 7.4

Determine O.N of Br in
 H_3BO_3

Solution:-

Let O.N of B = n

$$\text{then } 3(+1) + n + 3(-2) = 0$$

$$+3 + n - 6 = 0$$

$$n - 3 = 0$$

$$n = 3$$

EXAMPLE 7.5

Determine O.N of C in CO_3^{2-}

Solution:-

Let O.N of C = x

$$x + 3(-2) = -2$$

$$x - 6 = -2$$

$$x = -2 + 6$$

$$x = 4$$

Concept Assessment 7.3

One major problem of air pollution is the formation of acid-rain air pollutants. Such as SO_2 & NO_2 combine with O_2 & H_2O vapours in the air to form H_2SO_4 & HNO_3 .

i) determine the O.N of N in HNO_3

ii) S in SO_2 & H_2SO_4 .

Solution:-

i) let oxidation no of N = x

$$\text{then} = 1(+1) + x + 3(-2) = 0$$

$$1 + x + (-6) = 0$$

$$-5 + x = 0$$

$$\boxed{x = 5}$$

ii) let oxidation no of S = x in SO_2

$$\text{then} = x + 2(-2) = 0$$

$$x + (-4) = 0$$

$$\boxed{x = 4}$$

iii) let O.N of S = x in H_2SO_4

$$\text{then} = 2(+1) + x + 4(-2) = 0$$

$$2 + x + (-8) = 0$$

$$-6 + x = 0$$

$$\boxed{x = 6}$$

Concept Assessment 7.4

Determine the oxidation State of

i) S in Sulphate ion, SO_4^{-2}

Let O.S of S = x

$$\text{then } x + 4(-2) = -2$$

$$x + (-8) = -2$$

$$x = -2 + 8$$

$$x = 6$$

ii) P in phosphate ion, PO_4^{-3}

Let, O.S of P = x

$$\text{then } x + 4(-2) = -3$$

$$x + (-8) = -3$$

$$x = -3 + 8$$

$$x = 5$$

iii) N in ammonium ion, NH_4^{+1}

Let N O.S = x

$$\text{then } x + 4(+1) = +1$$

$$x + 4 = +1$$

$$x = +1 - 4$$

$$x = -3$$

Topic:-

7.3

"Formula of an ionic Compounds"

Question no 1:- how can we Calculate the formula of an ionic Compounds?

Answer:-

Determine the Simplest whole no. ratio of Cations to anions that result in a neutral Compound, ionic Compounds are electrically neutral.

Example:-

i) Magnesium-oxide (MgO)

Step 1 = Identify the ions & their charges magnesium ion (Mg^{2+}) oxide ion (O^{2-}).

Step 2 = determine the charge - M has a charge of 2 & a oxide has charge of -2. Thus, One M ion is require for each oxide ion.

Step 3 = Write the formula - MgO

Topic:-

7.4

"Oxidizing & reducing agents"

Question No 1:- What are oxidizing & reducing agents?

Answer:-

Oxidizing agents:-

"An oxidizing agent is that cause other substance to oxidize by taking electron's from it".

It is often called electron acceptor, the oxidizing agent themselves are reduced in the process (than gain electron).

Example:- O_2 , H_2O_2 etc.

Reducing agents:-

"A reducing agent is a substance that cause other substance to be reduced by donating electron to it".

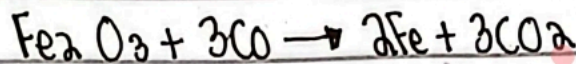
It is often called electron donor, Reducing agent themselves are oxidized in process

Example:- H_2 , CO

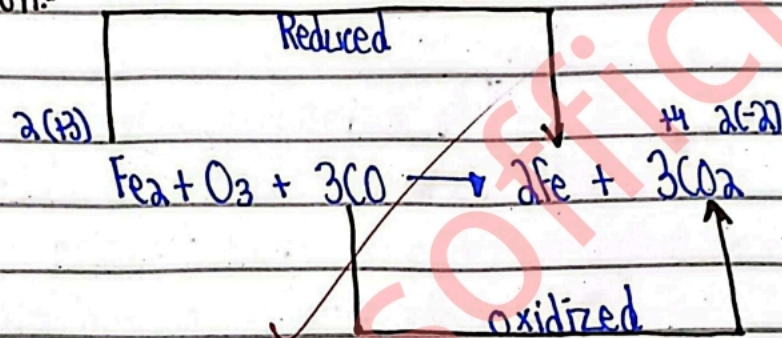
Topic:- 7.4.2

Question:-

Identify the oxidizing & reducing agents in the following reaction?



Solution:-

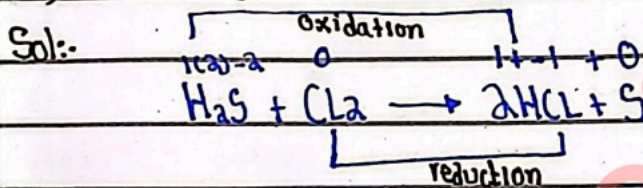


In this reaction O.N of Fe decrease from "+3" to "0" So it is reduced & Fe_2O_3 is oxidizing agent whereas O.N of C increase from "+2" to "+4" So it is oxidized & "CO" is reducing agent.

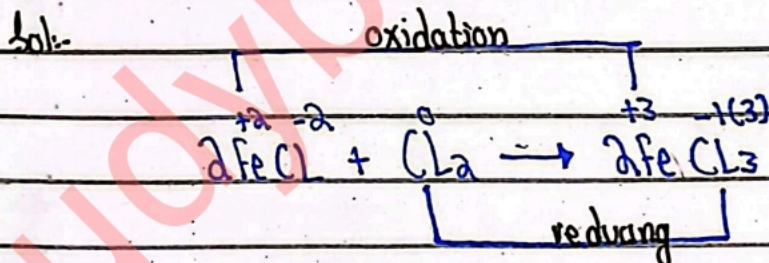
EXERCISE Question NO 9 part b

Part b:~

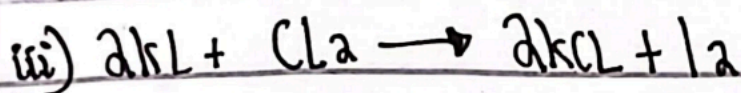
Identify oxidizing & reducing agents in the following reactions.



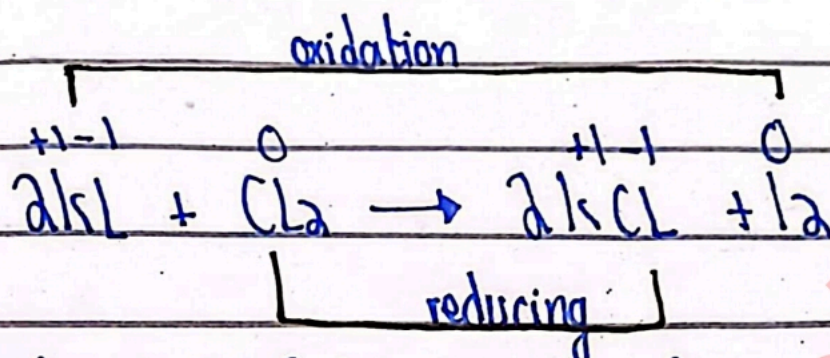
In this reaction 'H' is losing O.N from 2 to 1 So it is "oxidizing" and it is "reducing agent" whereas O.N at Cl reduce from 0 to -1 So it is "reducing" and it is "oxidizing agent"



In this reaction "Fe" O.N is increasing from +2 to +3 So it is oxidizing and it is reducing agent whereas O.N increase from 0 to +3 So it is reduced & it is oxidizing agent.



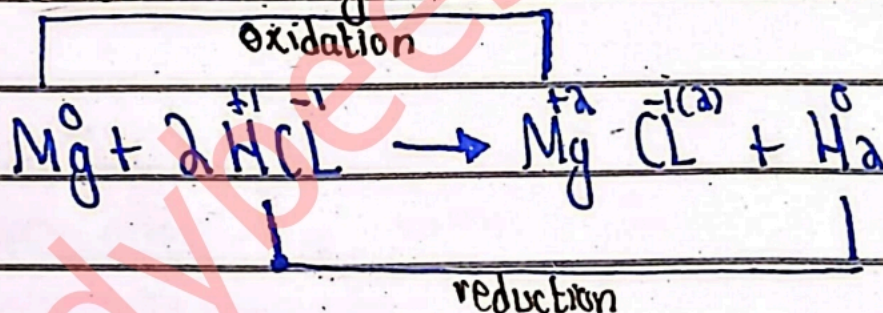
Sol



In this reaction O.N of I is increasing from -1 to 0

So it's "oxidizing" So it "reducing agent" whereas,

O.N of all Cl is "reducing" from 0 to -1, So it is "oxidizing agent"



In this reaction O.N of Mg increase from 0 to +2

So it is "oxidized" So it is "reducing agent" where

as, O.N of H decrease from +1 to 0 So it is reduced

So it is "oxidizing agent"

Topic:- 7.5

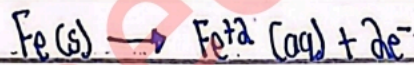
"Corrosion & its prevention"

Question No 1:- What is Corrosion?

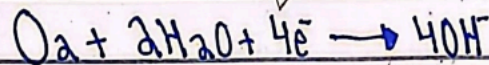
Answer:- "In this reaction a metal reacts with oxygen & moisture in the atmosphere"

Corrosion Converts refined metals to the more stable metal oxide. It can cause significant damage to structures, vehicles, & equipments.

Most familiar eg of corrosion is the formation of rust on iron. O_2 & H_2O are necessary for iron to rust. Corrosion is an oxidation-reduction reaction. A region of metal surface that has relatively less moisture, acts as anode; Will Fe oxidize in this region?



Another region on the surface that has relatively more moisture acts as cathode. The electrons released in the oxidation process reduce atmospheric oxygen to hydroxyl ions.



The Fe^{2+} ions formed at the anodic regions flow to the cathodic regions & through the moisture on the surface. Here Fe^{2+} ions further react with O_2 to form rust, $Fe_2O_3 \cdot xH_2O$.

Question No 2:- What are the prevention of Corrosion?

Answer:- Following are the methods that help to prevent Corrosion.

i) Coating with paints:- Corrosion can be prevented by applying protective coating such as paint or epoxy creating a barrier b/w metal surface & its environment.

ii) Alloying:- The tendency of iron to oxidise can be greatly reduced by alloying it with other metals
For eg:- adding chromium to form stainless steel which is highly resistant to corrosion.

iii) Coating with a thin layer of another metal:- Metals that readily corrode can be protected by coating. This can be done by:-

a) Tinning.

b) Galvanizing.

c) Electroplating.

(a) Tinning:- In this the process of tin plating, clean iron sheet is dipped in a bath of molten tin. It is then passed through hot pair of rollers. Tin protects iron effectively, since it is very stable.

(b) Galvanizing (Coating with Zinc):- The process of galvanizing consist of dipping a clean iron sheet is rolled into zinc bath & cooled.

(c) Electroplating:- In electroplating an electrolytic process is used to deposit one metal on another metal.

iv) Cathodic protection:- also known as sacrificial protection is the process in which the metal to be protected from corrosion is made cathode & connected to an active metal such as magnesium or aluminium. These metals are more active than iron they act as anode. The more active metal oxidizes itself & saves iron from corrosion. Cathodic protection is employed to prevent iron & steel structure such as pipes, tanks, oil rigs etc.

EXERCISE QUESTIONS

Question no. 2: short Q/A.

ii) What is the oxidation state of "Cr" in chromic acid (H_2CrO_4)?

Ans:- let the O.N of Cr = x

$$= 2(+1) + x + 4(-2) = 0$$

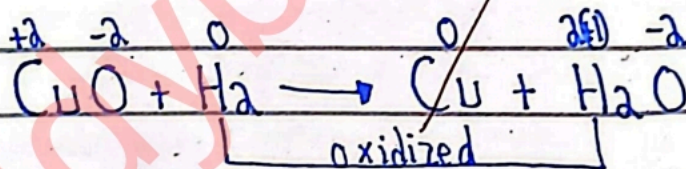
$$= 2 + x - 8 = 0$$

$$= x - 6 = 0$$

$$= x = 6$$

iii) Identify reducing agent in the following reaction.
reaction:- $CuO + H_2 \rightarrow Cu + H_2O$

Ans:-



As O.N of 'H' increase from '0' to +1. So it is oxidized & H_2 is reducing agent.

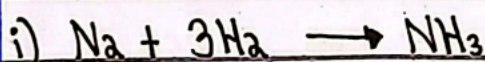
iv) Why tin plated steel is used to make food cans?

Ans:- Tin plated steel prevents corrosion protect food from reaction. it also prevent rust.

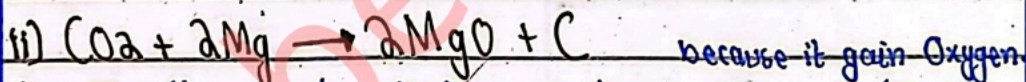
vi) Give 1 example from daily life which involved oxidation reduction reaction?

Ans:- One example of oxidation and reduction in our daily life is the rusting of iron. Iron react with oxygen in the presence of moisture forming iron-oxide. In this reaction iron lose electron (oxidation) while oxygen gain electron (reduction).

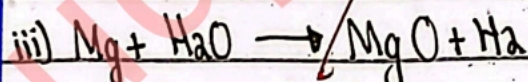
Question no 6:- state the substance which are oxidized or reduced. Give reason for your answer.



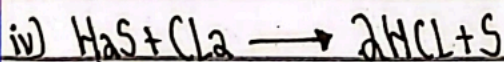
Ans:- "N" is gaining "H" so it is reduced, whereas H is oxidized.



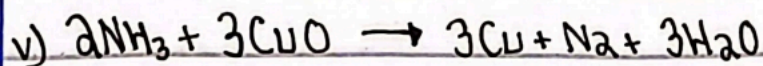
Ans In this reaction "Mg" is oxidized where as "C" is reduced. because it loses oxygen.



Ans:- "Mg" is oxidized, & H is "reduced".
it gain oxygen because H lose oxygen



Ans:- "H₂S" is oxidized, and "Cl₂" is "reduced".
because it lose "S" because it gain "H"



Ans:-

In this reaction "CuO" is losing oxygen so it is reduced whereas "N" is gaining so it is oxidized.

Question no 7 part b :- find the oxidation state of nitrogen in the following

i) NO_2

let O.N of "N" = x

$$x + 2(-2) = 0$$

$$x - 4 = 0$$

$$\boxed{x = 4}$$

$$x = 6/2$$

$$\boxed{x = 3}$$

ii) Na_2O

let O.N of "N" = x

$$2x + (-2) = 0$$

$$2x - 2 = 0$$

$$2x = 2$$

$$x = 2/2$$

$$\boxed{x = 1}$$

iv) HNO_3

let O.N of "N" = x

$$1 + x + 3(-2) = 0$$

$$1 + x - 6 = 0$$

$$x - 5 = 0$$

$$\boxed{x = 5}$$

iii) N_2O_3

let O.N of "N" = x

$$2x + 3(-2) = 0$$

$$2x - 6 = 0$$

$$2x = 6$$

Question no 2: Find oxidation state of S in the following compounds.

i) ~~NOH~~ H_2S

let O.N of S = x

$$\cancel{1(+1)} + x = 0$$

$$2(+1) + x = 0$$

$$2 + x = 0$$

$$x = -2$$

ii) H_2SO_3

let O.N of S = x

$$2(+1) + x + 3(-2) = 0$$

$$2 + x - 6 = 0$$

$$x - 4 = 0$$

$$x = 4$$

$$\boxed{x = 4}$$

iii) $\text{Na}_2\text{S}_2\text{O}_3$

let O.N of S = x

$$2(+1) + 2x + 3(-2) = 0$$

$$2 + 2x - 6 = 0$$

$$2x - 4 = 0$$

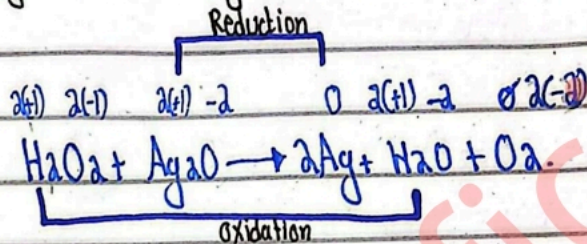
$$2x = 4$$

$$x = 2$$

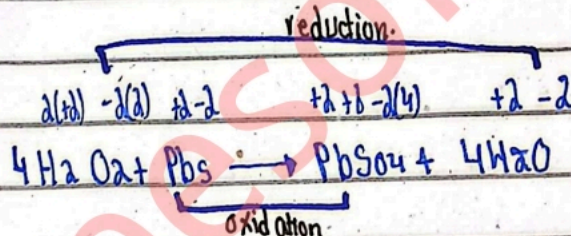
Question no 10:- Is hydrogen peroxide an oxidizing or reducing agent in these reactions. **Answer:-**



Sol:-



Sol:-



Question no 5:- Explain how the Food and beverage industries deal with Corrosion.

Answer:- The food and beverage deal with Corrosion by using materials that are resistant to Corrosion, Such as stainless Steel, aluminum, and tin-plated steel. These material help to prevent the Containers and equipment from rusting and ensure the safety and quality of the Food and beverages. Additionally, protective Coatings and packaging material are used to further reduce the risk of Corrosion.